

Date	01 October 2026
Team ID	SWUID2026355656
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Project Design Phase

Proposed Solution Template

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Managing library inventory manually is time-consuming, prone to errors, and makes it difficult for students to find books or track due dates, leading to lost assets and unpaid fines.
2.	Idea / Solution description	A Smart Library Management System (LMS) with a web portal for students to search books, reserve copies, and track due dates, alongside an admin dashboard for librarians to manage inventory, issue/return books, and automate fine calculations.
3.	Novelty / Uniqueness	Integration of automated email reminders for due dates, a machine learning-based book recommendation engine based on user reading history, and real-time dashboard analytics for library usage.
4.	Social Impact / Customer Satisfaction	Enhances the student academic experience by saving time, encourages reading through easy access, and significantly reduces the administrative burden on library staff.

5.	Business Model (Revenue Model)	Subscription-based SaaS model for educational institutions and public libraries, tiered by the number of active users and books. Additional revenue from premium analytics features.
6.	Scalability of the Solution	Cloud-hosted architecture (using AWS/Kubernetes) allows the system to easily scale from a small departmental